Prayas JEE 2025

Maths

Sequence and Series

DPP: 5

Q1 If sum of an infinite geometric series is $\frac{4}{3}$ and its 1^{st} term is $\frac{3}{4}$, then its common ratio is

(A) $\frac{7}{16}$ (B) $\frac{9}{16}$
(C) $\frac{1}{9}$ (D) $\frac{7}{9}$

Q2 If a, b and c are in $\mathrm{G.P.}$, then $\frac{b-a}{b-c}+\frac{b+a}{b+c}=$

(A) b^2-c^2
(B) ac
(C) ab
(D) 0

Q3 If the first term of a GP is 5 and common ratio is -5, then which of the following term is 3125?

(A) 6^{th}
(B) 5^{th}
(C) 7^{th}
(D) 8^{th}

Q4 The 6^{th} term of a GP is 32 and its 8^{th} term is 128, then the common ratio of the GP is

(A) -1 (B) 2
(C) 4 (D) -4

Q5 If $x,(2x+2),(3x+3)$ are in GP, then the fourth term is

(A) 27 (B) -27
(C) 13.5 (D) -13.5

Q6 If the p^{th}, q^{th} and r^{th} terms of a GP are a, b, c then $\left(\frac{c}{b}\right)^p\left(\frac{b}{a}\right)^r\left(\frac{a}{c}\right)^q$ is equal to

(A) 1
(B) $a^p b^q c^r$
(C) $a^q b^r c^p$
(D) $a^r b^p c^q$

Q7 If p^{th}, q^{th} and r^{th} term of a GP are a, b, c respectively, then $a^{q-r}\cdot b^{r-p}\cdot c^{p-q}$ is equal to

(A) 0 (B) 1
(C) abc (D) pqr

Q8 Three numbers are in GP such that their sum is 38 and their product is 1728. The greatest number among them is

(A) 18 (B) 16
(C) 14 (D) None of these

Q9 If a, b and c are in GP, then

(A) a^2, b^2, c^2 are in GP
(B) $a^2(b+c), b^2(c+a), c^2(a+b)$ are in GP
(C) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in GP
(D) None of these

Answer Key

Q1	(A)
Q2	(D)
Q3	(B)
Q4	(B)
Q5	(D)
Q6	(A)
Q7	(B)
Q8	(A)
Q9	(A)